GCP-GCVE Audit Tracker

Brownfield Audit Checklist and Remediation Tracker

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Table of Contents

1. Purpose

2. How to use this tracker

3. Audit Tracker Table

4. Next Steps

5. References

1. Purpose

This document provides a structured audit tracker for assessing and remediating network gaps between Google Compute Engine (GCE) and Google Cloud VMware Engine (GCVE).

2. How to use this tracker

- Use the Status column to mark: Not Started, In Progress, Blocked, Complete.
- Assign an Owner for each row and update progress notes in your internal system.
- Use the Improvement column as the remediation action to implement.

3. Audit Tracker Table

Audit Area	Current Gap	Improvement	Status	Owner
Topology	Mesh Peering	Migrate to NCC Hub-and-Spoke	Not Started	
Internet Access	VMs with External IPs	Use Cloud NAT + IAP	Not Started	
Security	Allow-All FW Rules	Hierarchical FW Policies	Not Started	
Traffic Flow	Unencrypted traffic	Internal HTTPS LB + mTLS	Not Started	
Visibility	No traffic logs	Enable VPC Flow Logs + Insights	Not Started	
Hybrid Path	Single VPN Tunnel	HA VPN or Interconnect	Not Started	
Redundancy	Single Region	Global LB + Multi-Region GCVE	Not Started	
Routing	Static/Manual routes	Global Dynamic Routing with Cloud Router	Not Started	
DNS	No DNS peering	Configure Cloud DNS Peering	Not Started	
IPAM	Overlapping CIDRs	Reserve Non-Overlapping CIDRs	Not Started	
Firewall	IP-based rules	Service Account-based rules	Not Started	
Monitoring	No connectivity tests	Run Connectivity Tests (Network Intelligence Center)	Not Started	
Compliance	No VPC Service Controls	Enable VPC Service Controls	Not Started	

4. Next Steps

1. Perform CIDR discovery across on-prem, GCE, and GCVE.

2. Prioritize remediation items by business impact and risk.

3. Assign owners and set target completion dates.

4. Re-run connectivity tests after each major change.

5. References

https://cloud.google.com/vpc/docs/shared-vpc

https://cloud.google.com/vpc/docs/private-services-access

https://cloud.google.com/network-connectivity/docs/router

https://cloud.google.com/dns/docs/overview#peering

https://cloud.google.com/architecture/best-practices-vpc-design

https://cloud.google.com/vmware-engine/docs/best-practices-security

https://cloud.google.com/nat/docs/overview

https://cloud.google.com/iap/docs/using-tcp-forwarding

https://cloud.google.com/vpc/docs/flow-logs

https://cloud.google.com/network-intelligence-center/docs/connectivity-tests/concepts/overview